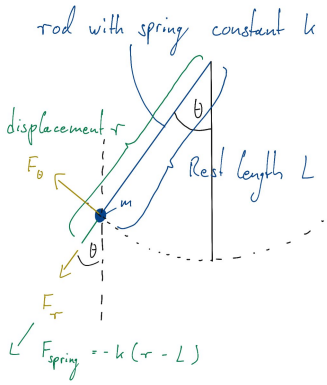
Newtonian Equations of Motion for an (In-)Elastic Pendulum in a Gravitational Field



Derivation from first principles in polar coordinates,
with numerical solution via `scipy.integrate.solve_ivp`
and real-time animation in `matplotlib`.

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Polar Coordinates

We transform from Cartesian (x, y) to polar (r, θ). The polar basis vectors are:

$$\hat{e}_r = \cos\theta\,\hat{e}_x + \sin\theta\,\hat{e}_y \tag{1}$$
$$\hat{e}_\theta = -\sin\theta\,\hat{e}_x + \cos\theta\,\hat{e}_y \tag{2}$$

Their time derivatives (since $\theta = \theta(t)$):

$$\dot{\hat{e}}_r = \dot{\theta}\,\hat{e}_\theta \tag{3}$$
$$\dot{\hat{e}}_\theta = -\dot{\theta}\,\hat{e}_r \tag{4}$$

Acceleration in Polar Coordinates

Starting from the position vector $\vec{r} = r\,\hat{e}_r$ and differentiating twice:

First derivative (velocity):

$$\dot{\vec{r}} = \dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta$$

Second derivative (acceleration):

$$\begin{aligned}\ddot{\vec{r}} &= \ddot{r}\,\hat{e}_r + \dot{r}\,\dot{\hat{e}}_r + \dot{r}\dot{\theta}\,\hat{e}_\theta + r\ddot{\theta}\,\hat{e}_\theta + r\dot{\theta}\,\dot{\hat{e}}_\theta \\ &= \ddot{r}\,\hat{e}_r + \dot{r}\dot{\theta}\,\hat{e}_\theta + \dot{r}\dot{\theta}\,\hat{e}_\theta + r\ddot{\theta}\,\hat{e}_\theta - r\dot{\theta}^2\,\hat{e}_r \\ &= \underbrace{(\ddot{r} - r\dot{\theta}^2)}_{a_r}\,\hat{e}_r + \underbrace{(r\ddot{\theta} + 2\dot{r}\dot{\theta})}_{a_\theta}\,\hat{e}_\theta \end{aligned} \tag{5}$$

The extra terms have physical meaning:

- $-r\dot{\theta}^2$ is the **centripetal acceleration**, pointing inward. A radial force (tension) is needed to sustain circular motion.
- $2\dot{r}\dot{\theta}$ is the **Coriolis acceleration**. It couples radial and angular motion: when $\dot{r} \neq 0$ and $\dot{\theta} \neq 0$, the angular velocity must adjust to conserve angular momentum.

Force Decomposition

Gravity mg acts vertically downward. Decomposed into polar components (with θ measured from the downward vertical):

$$F_r = mg\cos\theta \tag{6}$$
$$F_\theta = -mg\sin\theta \tag{7}$$

Applying Newton's second law $F = ma$ in each direction:

$$F_r = m(\ddot{r} - r\dot{\theta}^2) = mg\cos\theta \tag{8}$$
$$F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -mg\sin\theta \tag{9}$$

Case 1: Rigid Pendulum — $r(t) = L$

For a rigid rod: $r = L$ (constant), so $\dot{r} = 0$ and $\ddot{r} = 0$.

Radial equation (rod tension)

Let T denote the rod tension acting in the $-\hat{e}_r$ direction:

$$-mL\dot{\theta}^2 = mg\cos\theta - T$$

$$\boxed{T = mg\cos\theta + mL\dot{\theta}^2}$$

This is the sum of the gravitational component along the rod and the centripetal force.

Tangential equation (equation of motion)

The Coriolis term vanishes ($\dot{r} = 0$), leaving:

$$mL\ddot{\theta} = -mg\sin\theta$$

$$\boxed{\ddot{\theta} = -\frac{g}{L}\sin\theta}$$

This is the full nonlinear pendulum equation (no small-angle approximation). It is solved numerically with initial conditions $\theta(0)$ and $\dot{\theta}(0)$.

Case 2: Elastic Pendulum — $r(t)$ variable

The rigid rod is replaced by a spring with constant $k = EA/L$, where E is Young's modulus, A the cross-sectional area, and L the rest length. The spring force $-k(r - L)$ acts in the $\hat{e}_r$ direction.

The radial equation becomes:

$$m(\ddot{r} - r\dot{\theta}^2) = mg\cos\theta - k(r - L)$$

$$\boxed{\ddot{r} = g\cos\theta - \frac{k}{m}(r - L) + r\dot{\theta}^2}$$

The tangential equation:

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -mg\sin\theta$$

$$\boxed{\ddot{\theta} = -\frac{g}{r}\sin\theta - \frac{2\dot{r}\dot{\theta}}{r}}$$

These are **coupled** nonlinear ODEs — r appears in the θ-equation and θ appears in the r-equation. They can only be solved numerically, with initial conditions $r(0)$, $\dot{r}(0)$, $\theta(0)$, and $\dot{\theta}(0)$.

When Does the Pendulum Break?

The stress in the rod follows from Hooke's law with $k = EA/L$:

$$\sigma = \frac{F}{A} = \frac{k(r-L)}{A} = E \cdot \frac{r-L}{L}$$

The term $(r-L)/L$ is the **relative elongation** (strain). The material breaks when this stress exceeds the material-dependent **ultimate tensile strength** σ_{bruch}:

$$\sigma_{\text{bruch}} < E \cdot \frac{r-L}{L}$$

Rearranging for the critical extension:

$$\boxed{\Delta r_{\text{bruch}} := r - L = \frac{\sigma_{\text{bruch}} \cdot L}{E}}$$

This is the elongation of the pendulum rod at which it breaks.